SENSOR SIGNAL CONDITIONING TO SQUARE SIGNAL

Presented by

Nurmukhammed Polat

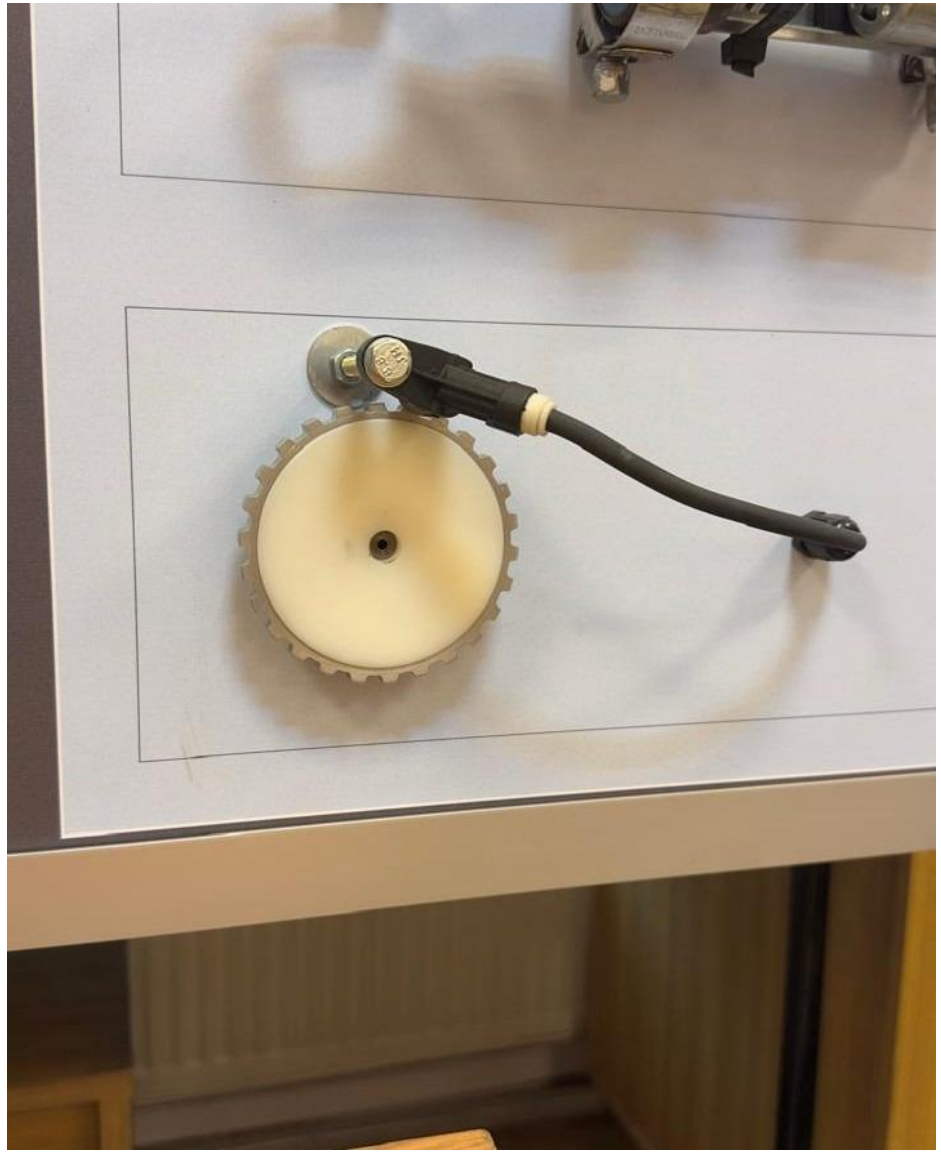
FINAL CIRCUIT SCHEMATIC

Rectifying diode bridge

Resistive voltage divider

Common collector stage – Emitter voltage follower

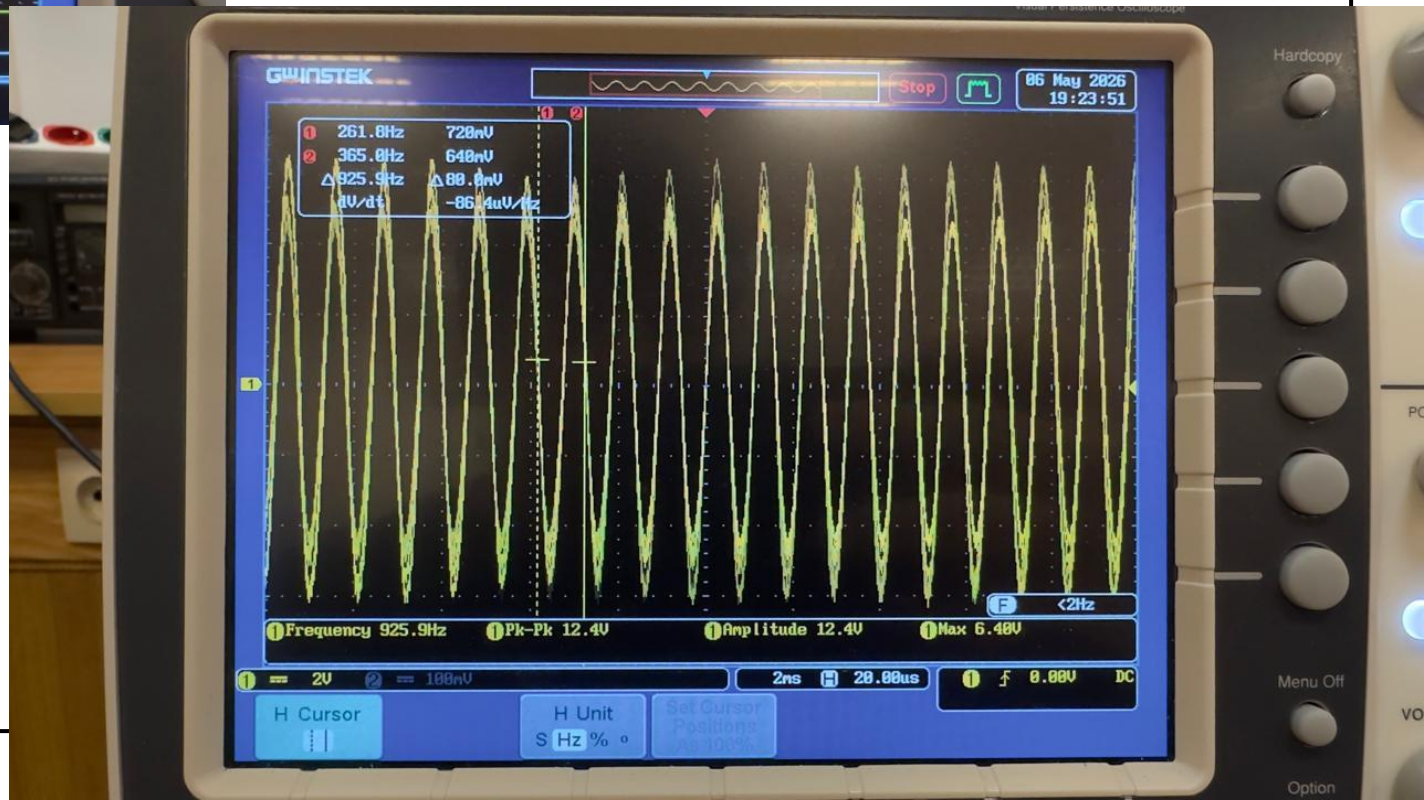
Common emitter stage



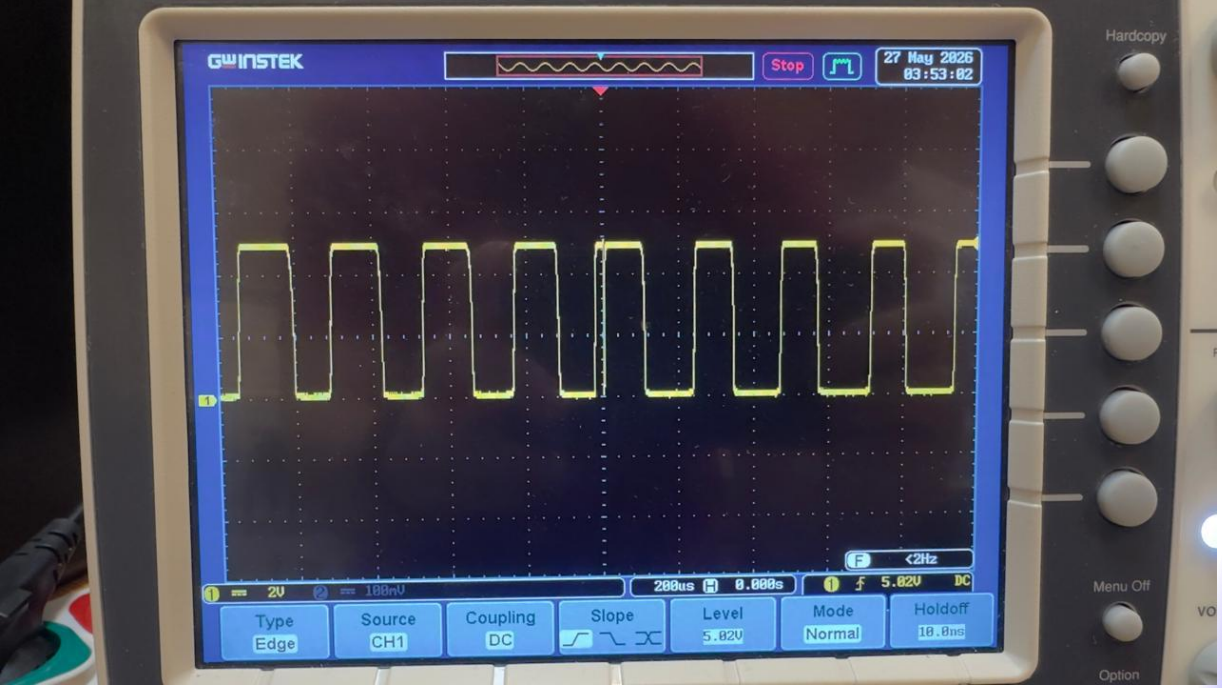
INPUT WAVEFORMS



From 6V to 24V peak to peak centering around 0V

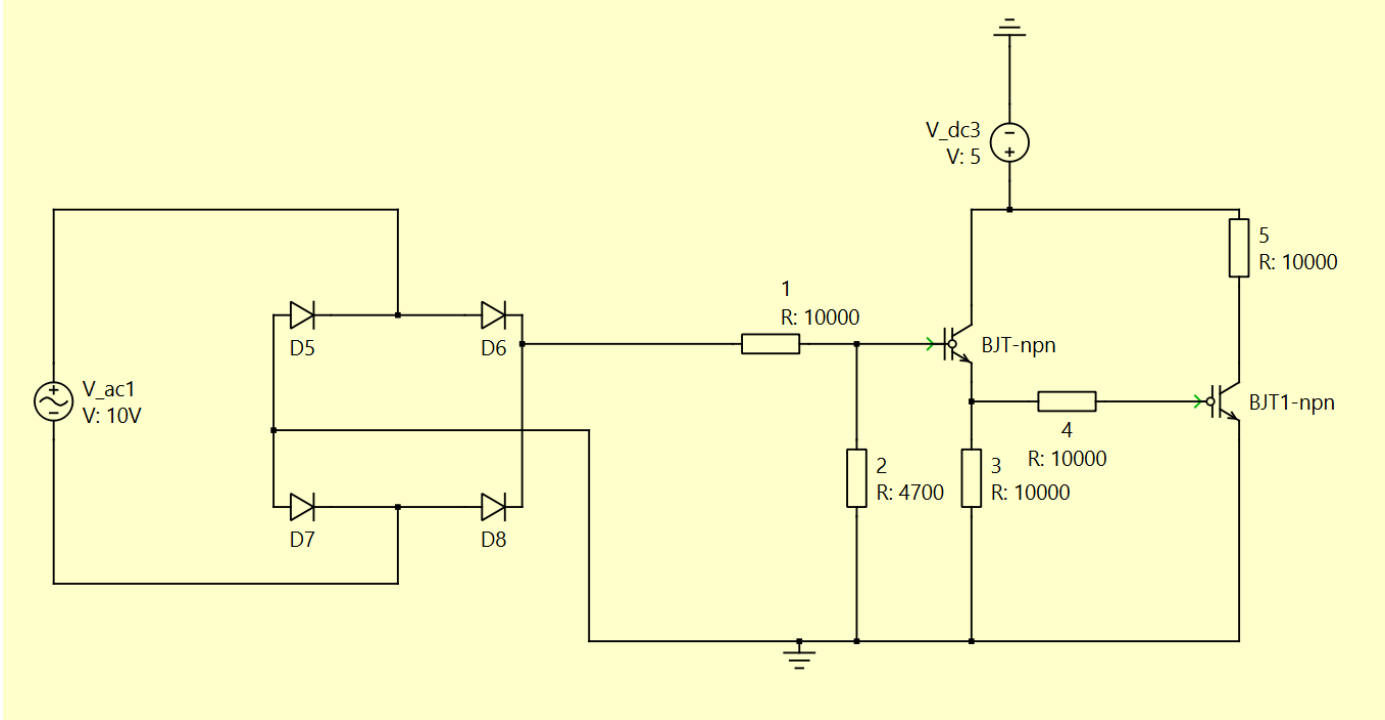


SOLUTION



WAVEFORM WE WANT – SQUARE WAVE

FINAL SOLUTION BUILT

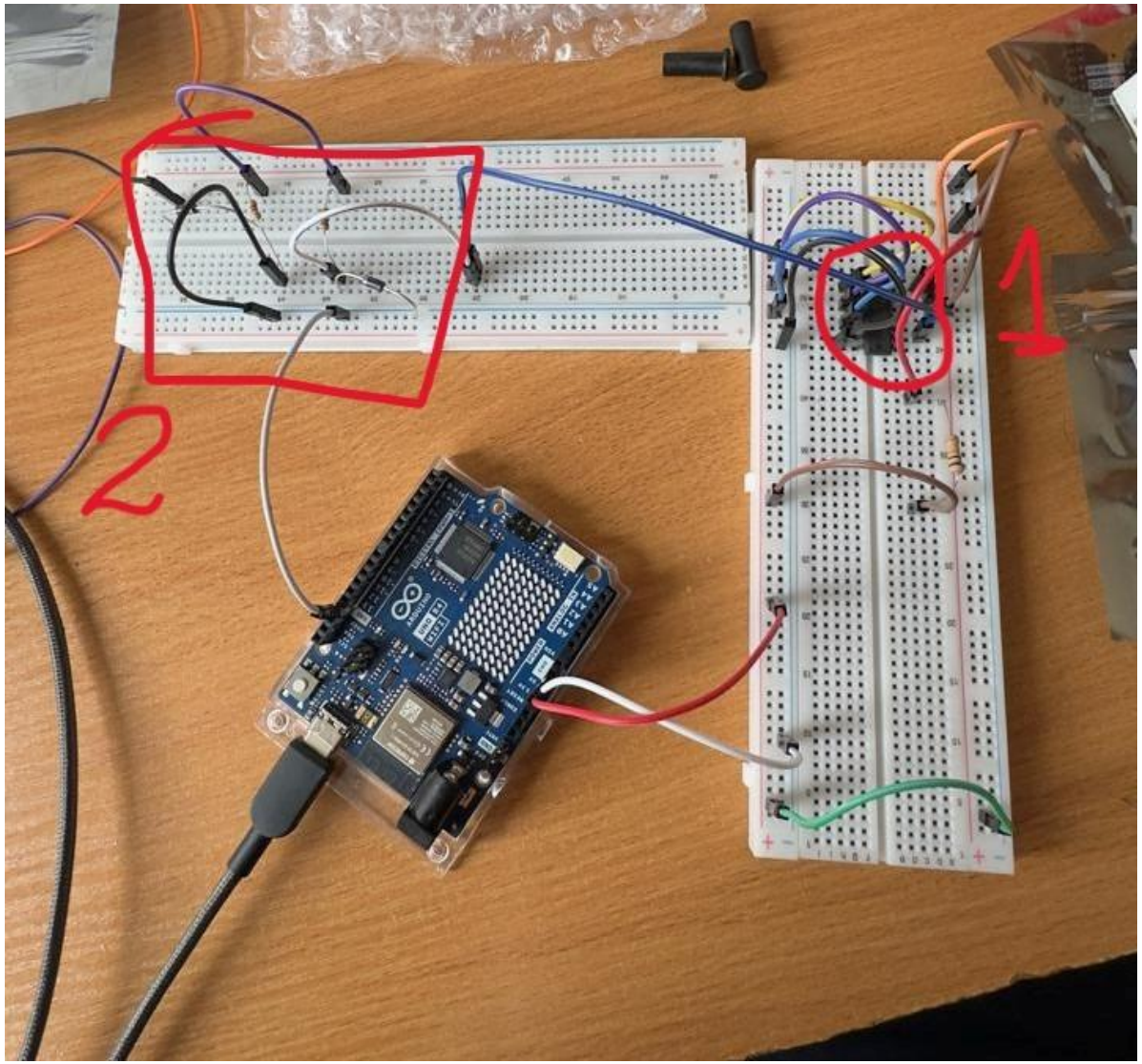


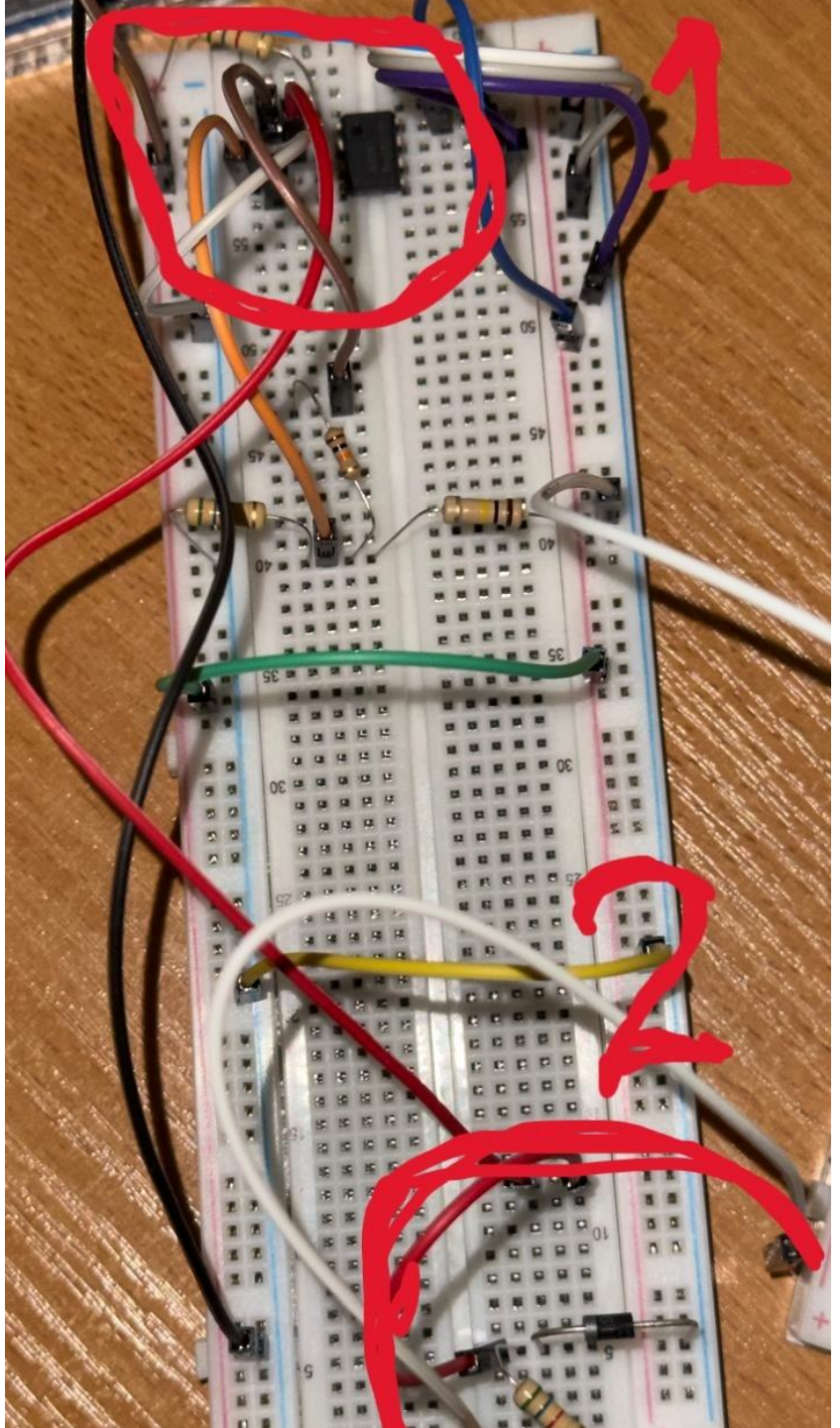
SUCCESSFUL hardware

Diodes – 1N4007 1A/1000V
BJT (npn) – BC547C

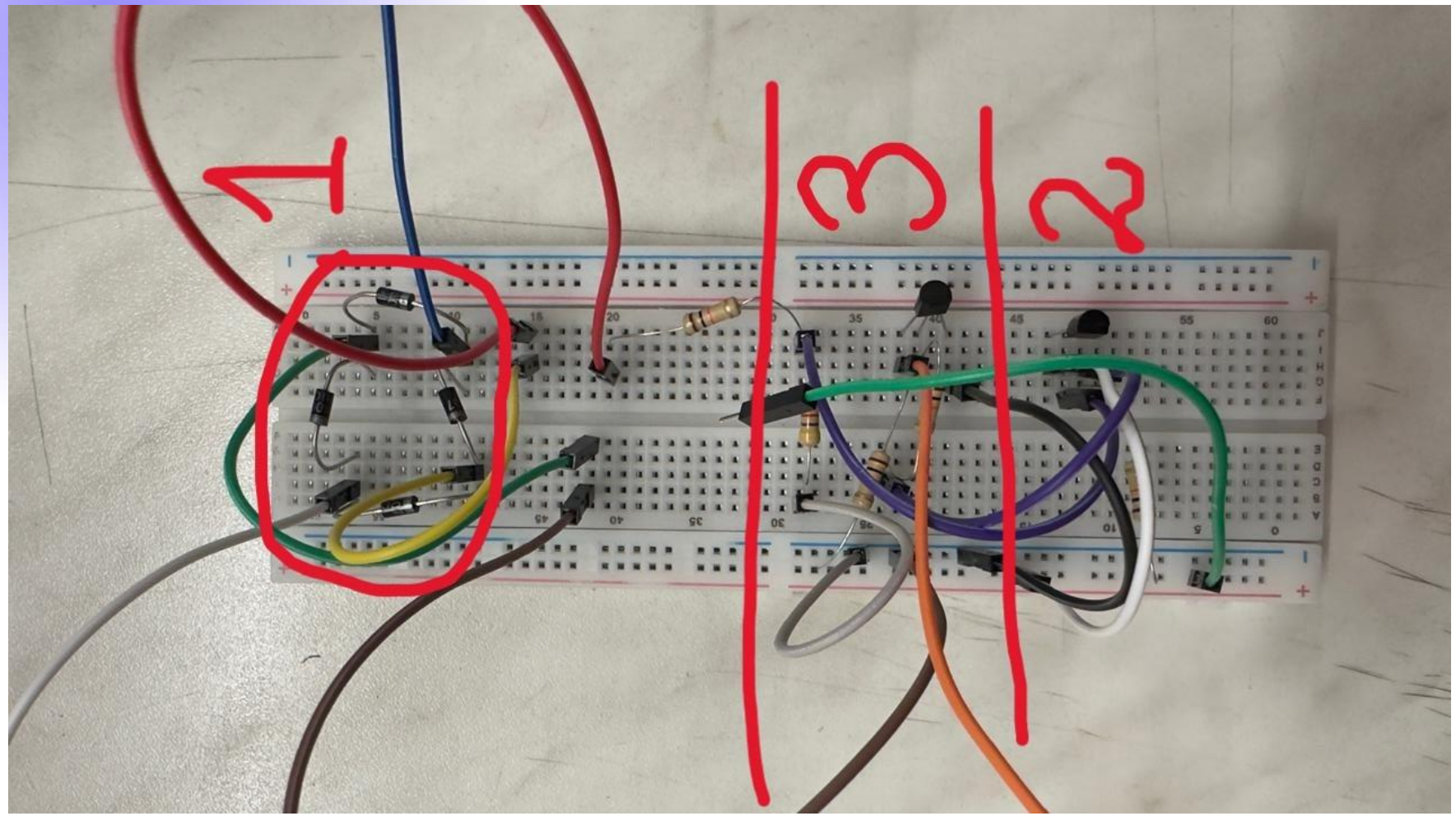
FAILED attempt hardware

Diodes – 1N4007 1A/1000V
Zener diodes – didnt find model. Maybe thats why circuits didnt work out
Comparator – LM393P
Schmitt trigger – SN74HC14N from Texas Ins


1
2


1
2

IMPLEMENTED CIRCUIT ON BREADBOARD

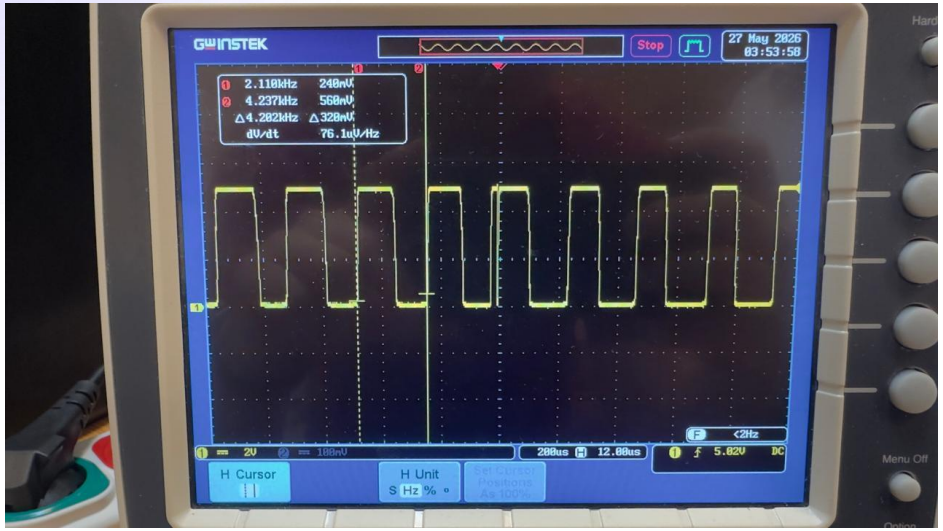


RESULTS AMPLITUDE AND FREQUENCY

Amplitude – 0V and 5V

Frequency – 4.2kHz/2 = 2.1kHz due to the diode rectification







Q & A